OpenSearch Project Vienna

Fall Meetup, October 13th 2026

Welcome Notes



Agenda

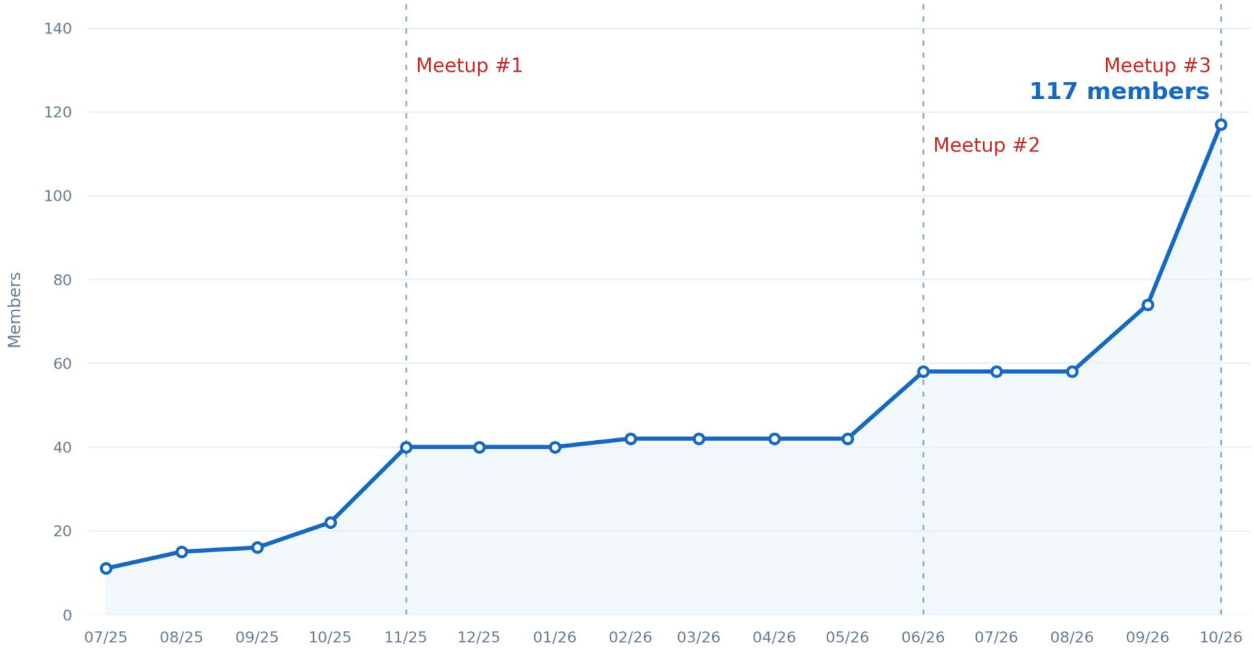
- Welcome Notes
- Talk 1: Developing Effective Security Analytics with OpenSearch
- Break (Drinks and Snacks)
- Talk 2: Inside OpenSearch’s AI toolkit: From Vector Search to Agents
- Networking (Drinks and Snacks)

Housekeeping & Logistics

- 📸 **Photos & Recording**
  - We'll record the talks and share them with the Meetup community.
  - We'll also take some photos for social media and the Meetup community
  - If you **don't want to appear** in photos or videos — just let us know
- 🤝 **Code of Conduct**
  - Let's be **respectful,**
  - **give space** to everyone,
  - **welcome** questions of our peers
  - And make it a **great time** for all of us

OpenSearch

OpenSearch Project Vienna - Timeline



Space for Future Ideas and Content

Talks, workshops,
hands-on,
tutorials, deep dives?

Let us know your ideas.

OpenSearch

OpenSearch Meetup - Around the World



OpenSearch

OpenSearch Meetup - Around the World



OpenSearch

Detection Engineering with OpenSearch

From security data to validated detections

OpenSearch + Sigma + Atomic Red Team™



$whoarewe

Paul Hager

Detection & Security Engineer | CERT.at

- Specialized in threat research and practical security detections
- Former Threat Researcher and Detection Engineer
- Experience in developing detection content for emerging threats

David Riepl

System Engineer | RISE

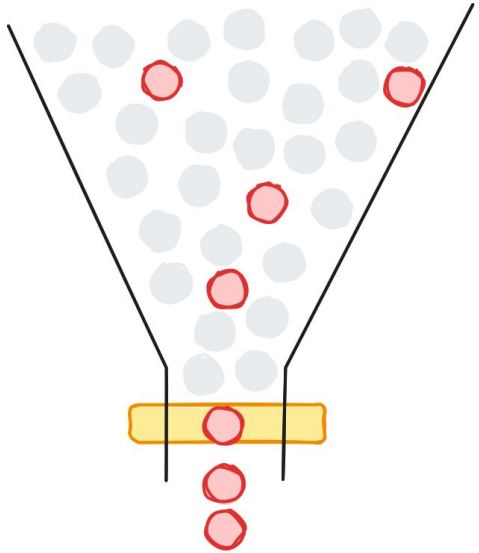
- Specialized in product development for log analysis and security monitoring
- Expertise in designing and implementing highly available systems
- Experience in security monitoring strategies for complex IT environments

Agenda

- The Problem: *Collecting Logs ≠ Detecting Attacks*
- Threat Model → Threat Research → Detection Engineering
- Validation: *Does it Actually Work?*
- OpenSearch as Detection Platform
- Demo
- Q & A

The Problem: Collecting Logs ≠ Detecting Attacks

- Huge amounts of (security) data
- Many different logs (formats, vendors, ...)
- Hundreds of possible rules
- False positives
- Unknown detection coverage
- Questions:
  - Where to start?
  - Does the detection actually work?
  - Where are my blindspots?

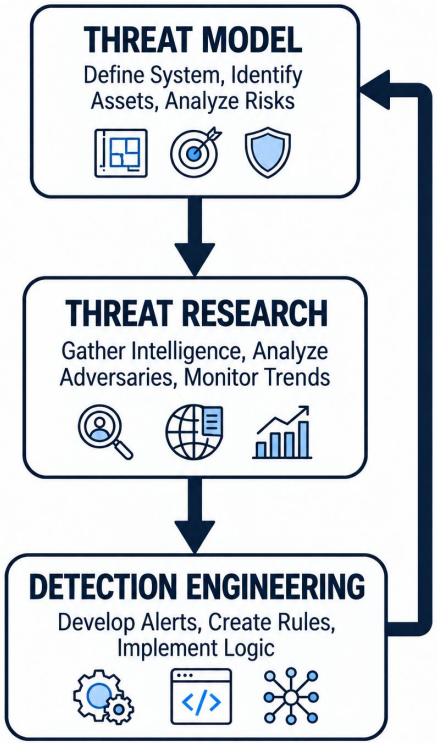


OpenSearch

Where to start?

- Threat Modeling:
  - Identify critical assets, attack surfaces, and potential threats.
- Threat Research:
  - Understand adversary behavior, techniques, and emerging threats.
- Detection Engineering:
  - Translate threats into actionable detection rules and alerts.
  - Continuous Validation: Test detections against real-world attack scenarios.

→ Detection Engineering is not a one-time activity, but a continuous, threat-driven improvement cycle.



OpenSearch

Where to start? - Threat Modelling

- What do you want to protect and why?
  - *“Crown Jewels”*
  - Prioritization
  - Gap analysis
- Useful Frameworks
  - STRIDE | Identify threat categories
  - MITRE ATT&CK | Map adversary tactics and techniques
  - PASTA | Risk-centric, business-driven threat modeling

“When gilding the safe, don't forget the front door.”



→ Threat modeling helps us build the *right detections*, rather than simply building *more detections*

Where to start? - Threat Research

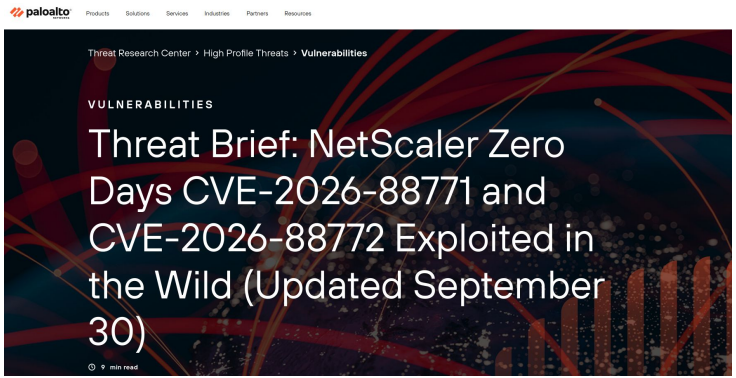
Goal: Identify and understand relevant threats

- Monitor threat intelligence sources
  - Security advisories, CVEs, research blogs, incident reports
  - Threat actor behavior and MITRE ATT&CK techniques
- Validate threats through simulation
  - Red / Purple Teaming
  - Atomic Red Team

→ turn information about real-world threats into actionable detection capabilities.

Example: NetScaler vulnerabilities

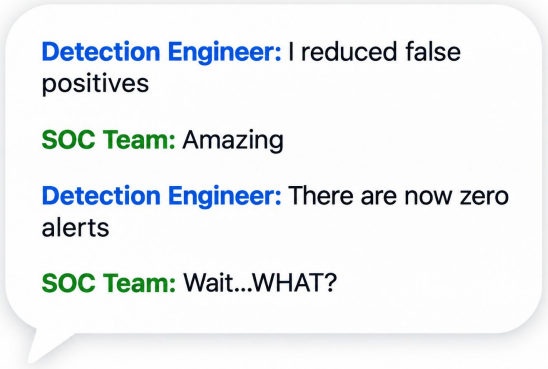
- Are we running affected systems?
- How are attackers exploiting them?
- What telemetry and detections do we need?



OpenSearch

Where to start? - Detection Engineering

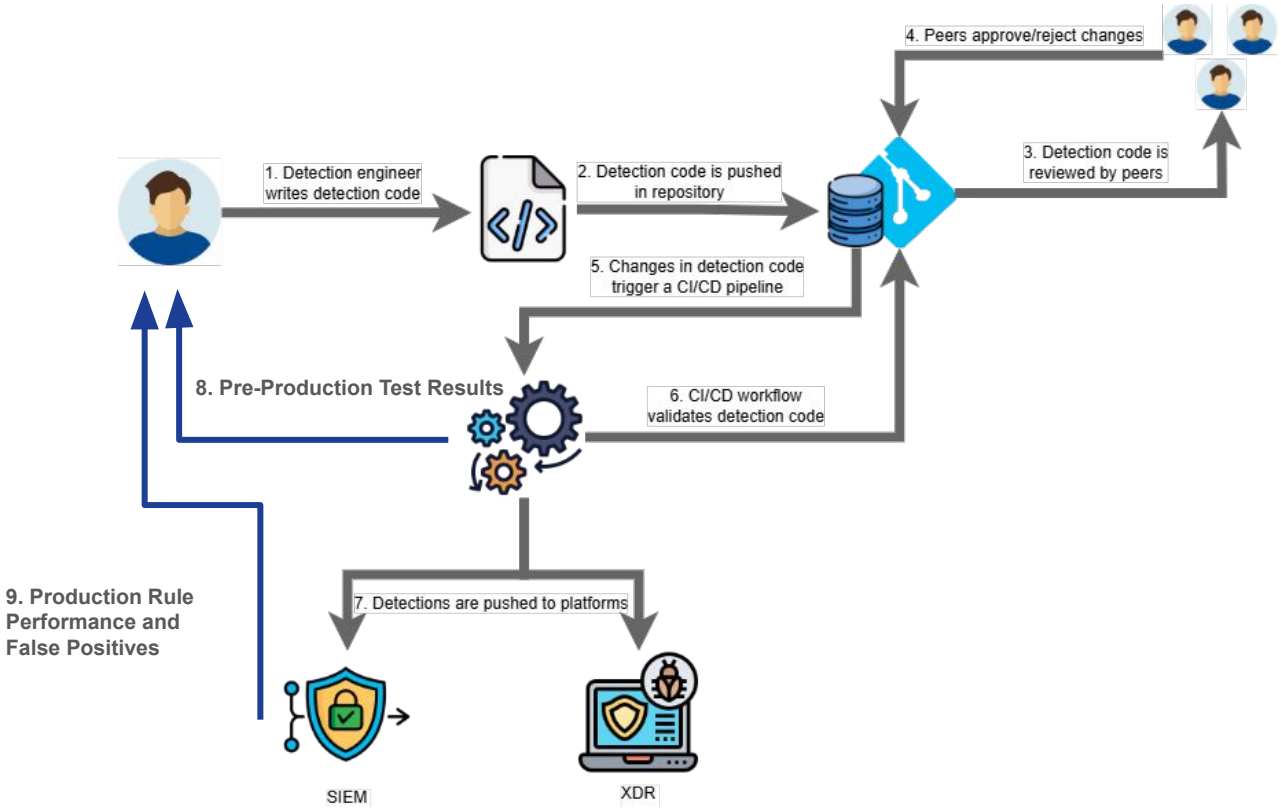
- Converts threat intelligence into actionable detections (e.g. Yara, Snort, **Sigma**, ...)
- “Systematically building and maintaining detections that identify malicious or suspicious activity within an organization’s environment” [1]
- Detection-as-Code follows software engineering best practices:
  - Version Control
  - Syntax checks
  - Regression tests
  - Code Reviews / Pull Requests



[1] https://www.rapid7.com/fundamentals/detection-engineering/

OpenSearch

Where to start? - Detection Engineering



https://blog.nviso.eu/2025/07/08/detection-engineering-practicing-detection-as-code-introduction-part-1/

Detection-as-Code with Sigma



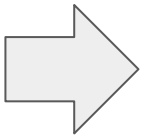
- “The shareable detection format for security professionals.”
- Write vendor and platform agnostic detection rules
- YAML
- Open Source
  - https://sigmahq.io/
  - https://sigconverter.io/

OpenSearch

Detection-as-Code with Sigma

```
sigma convert --without-pipeline -t lucene rule.yml
```

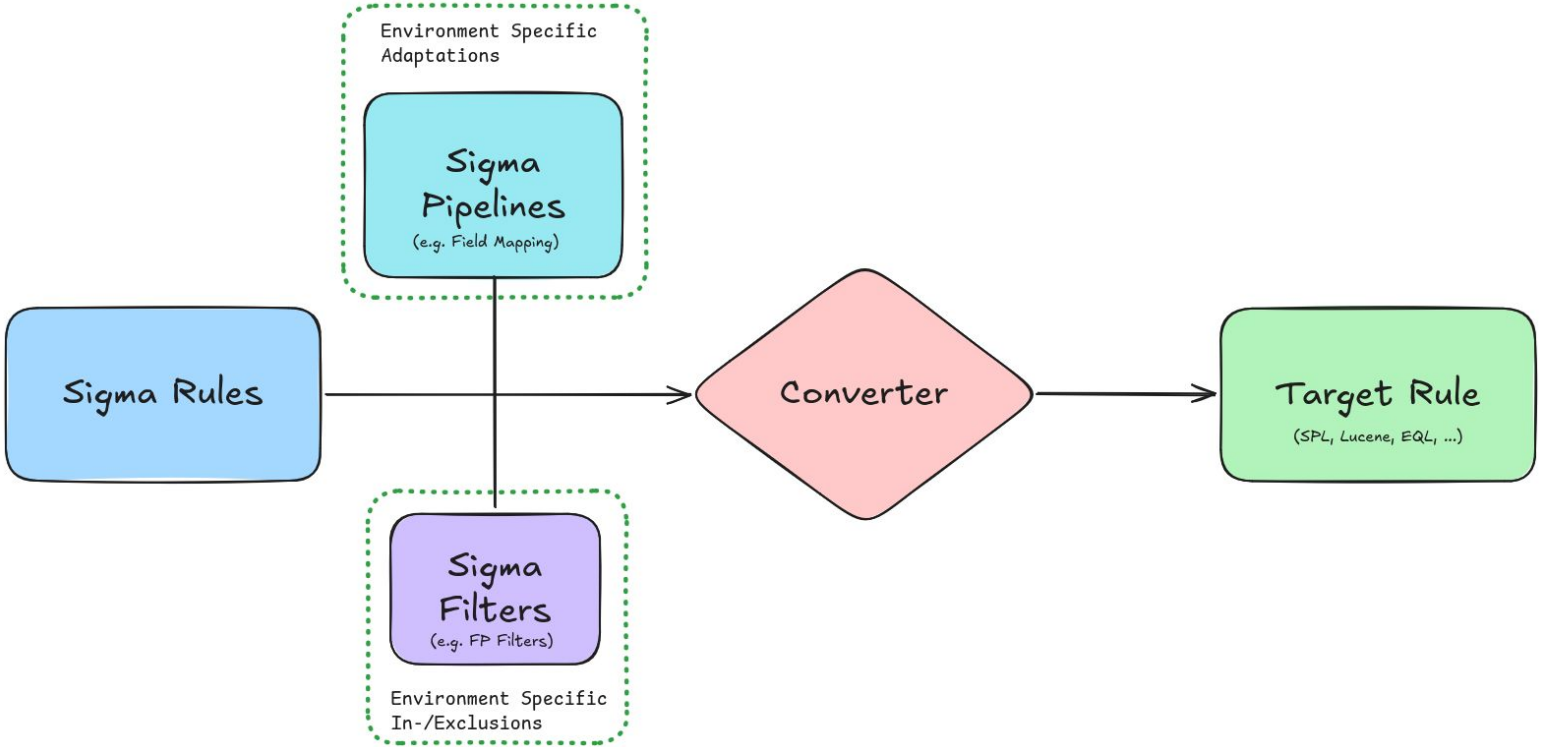
```yaml
title: AWS Root Credentials
description: Detects AWS root account usage
logsource:
    product: aws
    service: cloudtrail
detection:
    selection:
        userIdentity.type: Root
    filter:
        eventType: AwsServiceEvent
    condition: selection and not filter
level: medium
```



```
userIdentity.type:Root
AND (
    NOT eventType:AwsServiceEvent
)
```

OpenSearch

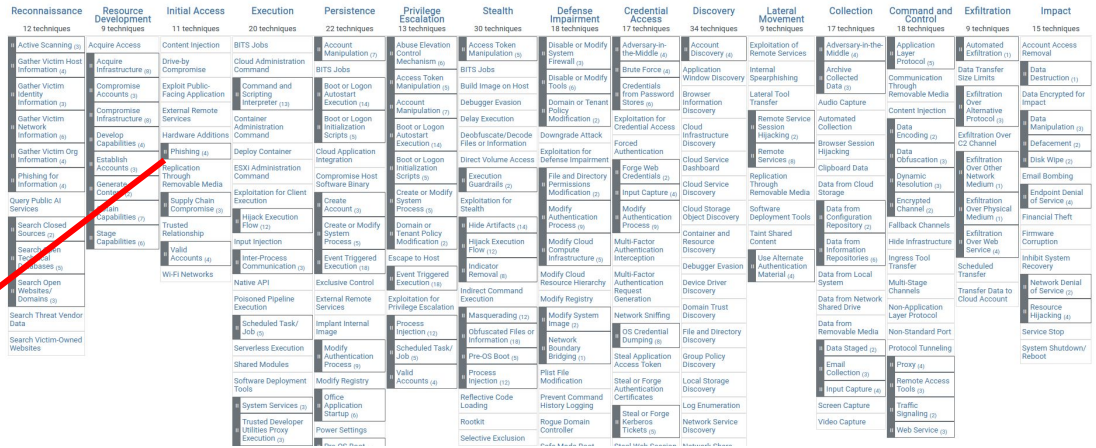
Detection-as-Code with Sigma



OpenSearch

But does the Detection Actually Work?

- MITRE ATT&CK®
  - "... a globally-accessible knowledge base of adversary tactics and techniques based on real-world observations." [1]
  - Taxonomy
    - Tactics & Techniques
  - Behaviors > IOCs
- e.g. "Phishing":
  - Initial Access (TA0001)
  - Phishing (T1566)





[1] https://attack.mitre.org/

OpenSearch

But Does the Detection Actually Work?

- Atomic Red Team™
  - “Atomic Red Team™ is a library of tests mapped to the MITRE ATT&CK® framework. Security teams can use Atomic Red Team to quickly, portably, and reproducibly test their environments.” [1]
- Coverage Test
  - Exec Atomic Red Team™ tests
  - Matches?
    - Yes → Profit
    - No → uh… back to detection engineering!

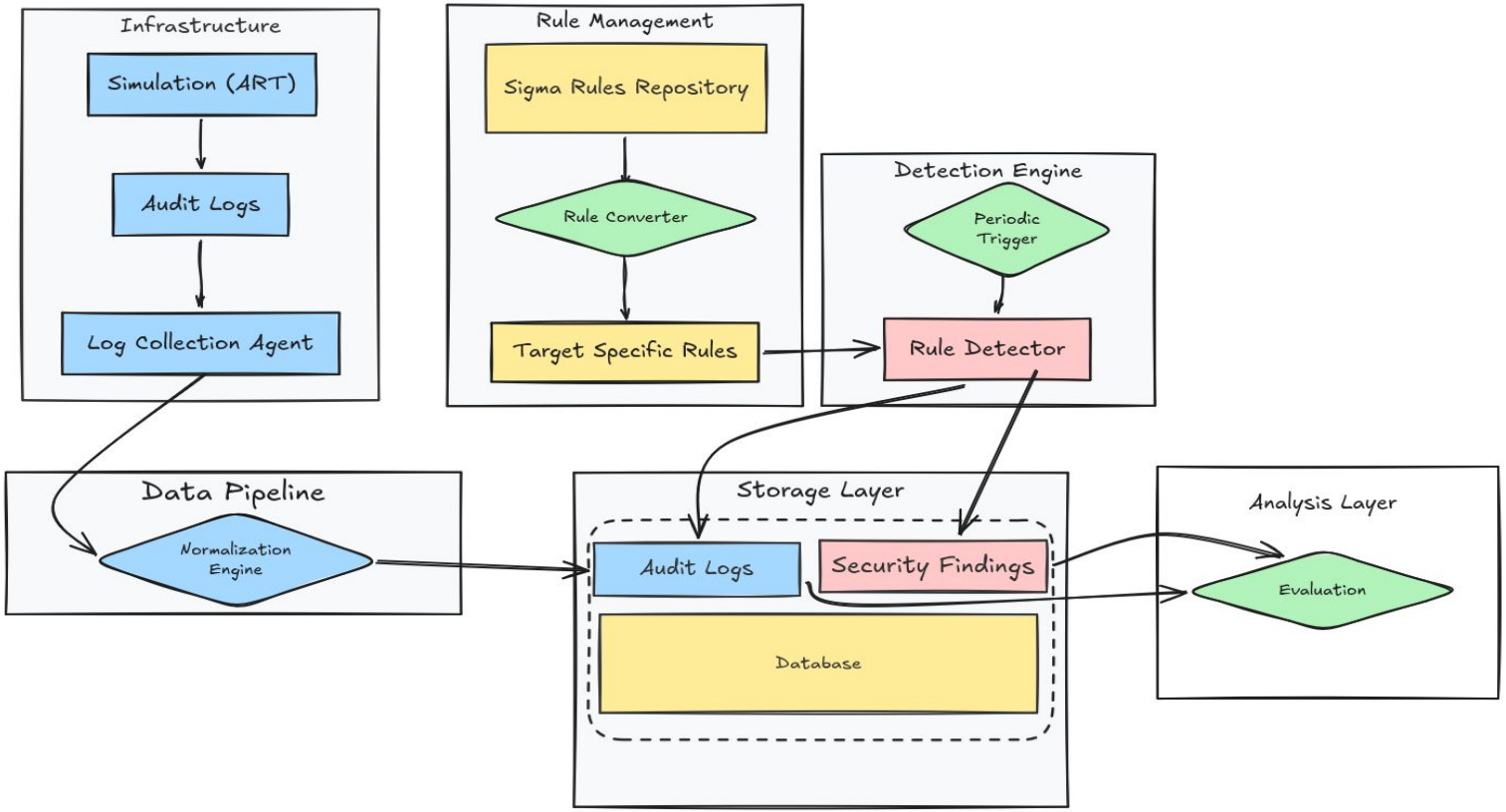
[1] https://github.com/redcanaryco/atomic-red-team

OpenSearch as the Detection Platform

- Why the OpenSearch ecosystem is a good place to start?
  - **Centralized Data Storage:** Ingest, normalize, and correlate logs and security telemetry from multiple sources.
  - **OpenSearch Security Analytics:**
    - Detection engine executes Sigma-based detection rules
    - Contains prepackaged rules for multiple security event log sources.
- Rule Management
  - Rule Repository
  - Field Mapping
- Simulation & Test Execution
  - Atomic Red Team Tests
  - Test and Simulation Environment

OpenSearch

OpenSearch as the Detection Platform



OpenSearch

Demo

Demo Setup - Technical Architecture

Goals:

- Quick Start Detection Engineering
- Show Coverage as Smoke Test Criteria for Rule Validation

- Containerized lab orchestrated with Docker Compose:
  - OpenSearch, OpenSearch Dashboards, Logstash, and Auditbeat
  - Inside a disposable vagrant VM
- **Attack Simulation:** Atomic Red Team tests run in separate disposable container
- **Log Collection:** Auditbeat captures Linux audit events and forwards them to Logstash.
- **Data Pipeline:** Logstash parses and normalizes events before indexing them into OpenSearch.
- **Detection Engine:** OpenSearch Security Analytics evaluates ingested events against Sigma-based detection rules and generates security findings.
- **Evaluation & Coverage:** OpenSearch Dashboards is used to correlate executed tests with security findings to assess detection coverage.

OpenSearch

Q & A